

Naive Bayes Foundations: A Bayes' Theorem Problem

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Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).



- **Builds on:** Conditional probability ([Note 82](#)).
- **Leads to:** Naive Bayes ([Note 87](#)).

1. Overview

Key point: Three machines make markers; a random marker turns out defective. Bayes' theorem, with the total probability rule for the evidence, gives the chance it came from each machine.

This Note works through a classic problem with Bayes' theorem. It also introduces the **law of total probability**, the usual way to compute the evidence $P(B)$ in the denominator. Naive Bayes computes its evidence the same way.

2. The problem

Key point: Production shares 20%, 30%, 50%; defect rates 5%, 3%, 1%.

A factory makes markers on three machines:

Machine	Share of all markers	Defect rate of that machine
M1	20%	5%
M2	30%	3%
M3	50%	1%

A marker is picked at random from a batch and turns out to be **defective**. What is the probability that it was made by M3?

3. Writing down what we know

Key point: The production shares are priors $P(M)$; the defect rates are likelihoods $P(D | M)$.

Let D be “the marker is defective”. In probability language:

- **Priors:** $P(M1) = 0.2$, $P(M2) = 0.3$, $P(M3) = 0.5$.
- **Likelihoods:** $P(D | M1) = 0.05$, $P(D | M2) = 0.03$, $P(D | M3) = 0.01$.

The question asks for $P(M3 | D)$. By Bayes' theorem,

$$P(M3 | D) = \frac{P(D | M3) \times P(M3)}{P(D)}$$

Everything on the right is known except $P(D)$, the overall probability that a marker is defective.

4. The evidence: total probability

Key point: A defective marker came from exactly one machine, so $P(D)$ is the sum of the three “defective and from machine i” probabilities.

A defective marker must have come from M1, M2 or M3, and from only one of them. These three cases are mutually exclusive (the mutually exclusive events Note), so their probabilities add:

$$P(D) = P(D \cap M1) + P(D \cap M2) + P(D \cap M3)$$

Each term is a likelihood times a prior, by $P(A \cap B) = P(B | A)P(A)$ from the Bayes' theorem proof:

$$P(D) = P(D | M1)P(M1) + P(D | M2)P(M2) + P(D | M3)P(M3)$$

This is the **law of total probability**. With numbers:

$$P(D) = 0.05 \times 0.2 + 0.03 \times 0.3 + 0.01 \times 0.5 = 0.010 + 0.009 + 0.005 = 0.024$$

So 2.4% of all markers are defective. Figure 1 shows the same calculation as a probability tree.

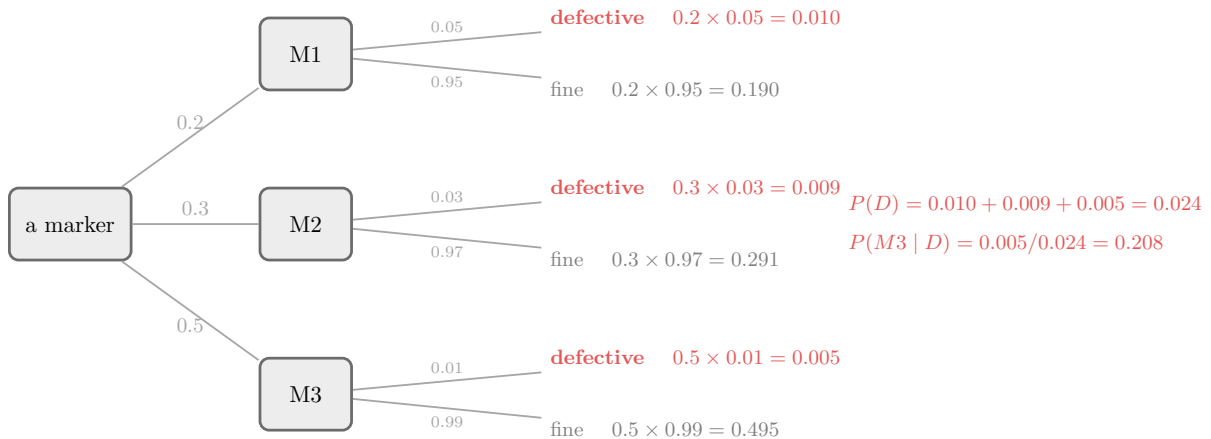


Figure 1: Probability tree: each path multiplies its branches

5. The answer

Key point: $P(M3 | D) = 0.005 / 0.024 = 0.208$. M3 makes half the markers but only about a fifth of the defective ones.

$$P(M3 | D) = \frac{0.01 \times 0.5}{0.024} = \frac{0.005}{0.024} = 0.208$$

The same calculation for every machine:

Machine	Prior $P(M)$	Joint $P(D \cap M)$	Posterior $P(M D)$
M1	0.20	0.010	0.417
M2	0.30	0.009	0.375
M3	0.50	0.005	0.208

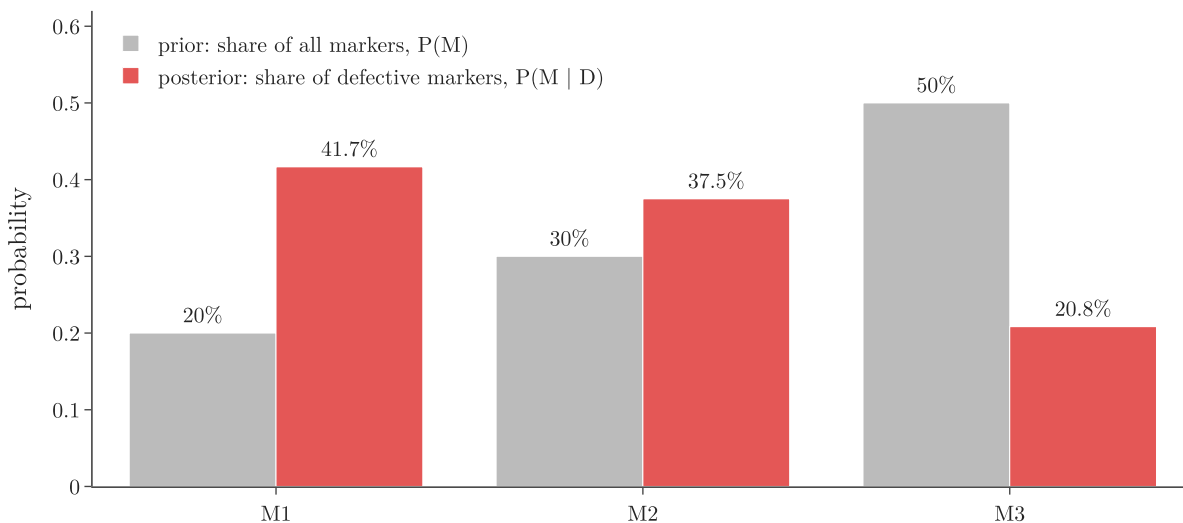


Figure 2: Share of all markers against share of defective markers

Seeing that the marker is defective changes the picture completely (Figure 2). Before, M3 was the most likely source (50%). After, it is the least likely (21%), because it rarely makes defects. M1 makes only 20% of markers but 42% of the defective ones.

The three posteriors add up to 1, as they must: the marker came from some machine. A simulation of a million markers gives $P(M3 | D) = 0.2095$, matching 0.208.

Python: The whole calculation.

```
prior = {"M1": 0.2, "M2": 0.3, "M3": 0.5}
p_defect = {"M1": 0.05, "M2": 0.03, "M3": 0.01}
```

```
joint = {m: prior[m] * p_defect[m] for m in prior}
p_D = sum(joint.values()) # 0.024
{m: joint[m] / p_D for m in prior} # 0.417, 0.375, 0.208
```

6. The pattern for Naive Bayes

Key point: For each class: prior $\times$ likelihood. Divide by their total. The largest posterior wins.

This is the exact procedure a Naive Bayes classifier follows, with machines replaced by classes and “defective” replaced by the observed inputs:

1. for every class, multiply its prior by the likelihood of the observation;
2. divide each product by their sum (the evidence) to get posteriors that add to 1;
3. predict the class with the largest posterior.

Since the evidence is the same for every class, step 2 does not change which class wins; it only turns the scores into probabilities.

7. Summary

- Priors: production shares; likelihoods: defect rates; posterior: which machine, given a defect.
- Law of total probability: $P(D) = \sum_i P(D | M_i)P(M_i)$.
- Answer: $P(M3 | D) = 0.208$; M1 is the most likely source (0.417).

8. Key terms

Term	Meaning
Law of total probability	$P(B) = \sum_i P(B A_i)P(A_i)$, when the A_i are mutually exclusive and cover every case
Probability tree	A diagram in which each path multiplies the probabilities along its branches
Joint probability	The probability that two events happen together, $P(A \cap B)$